

CPU searching

Williamette or Northwood, Athlon or Duron, the quest is neverending...

Over the past few months processors have gone through significant speed gains and attractive price cuts. There's a lot of buzz with the launch of new CPUs such as the Intel P4 2.4 GHz and the AMD Athlon XP 2100+ chips, while veterans such as the Socket 423-based P4 1.4/1.5 GHz (Williamette), the Athlon 1.4 GHz (Thunderbird) and the older Durons are slowly fading away.

My mission is crystal clear—To buy a good processor in less than Rs 8,000.

At my first stop, I state my budget and ask for a recommendation. The dealer suggests that a P4 1.6 GHz (Socket 478) is now available for as low as Rs 8,500. Just a little out of my budget, but an enticing option nonetheless. Just to be doubly sure, I ask the dealer to confirm whether it's a Northwood processor with 512 KB of L2 cache. The packaging mentions that the CPU has only 256 KB of L2 cache—it's the outdated Williamette core! You need to be careful as there are plenty of older Williamette core-based P4 processors in the market. While these CPUs support the latest Socket 478-pin format, they contain only 256 KB of L2 cache. The performance of these processors is about 10 per cent less than their Northwood counterparts. The older Williamette (Socket 423) processors with speeds up to 1.5 GHz can also be bought for a bargain price. If you're not thinking of upgrading in the next two years or so, then a cheap Socket 423 solution makes sense. Socket 423 motherboards have since long been out of fashion and many dealers will be ready to sell them at a discount. This makes them an attractive upgrade option.

I move on to the next dealer, this time shifting gears and inquiring about a P-III

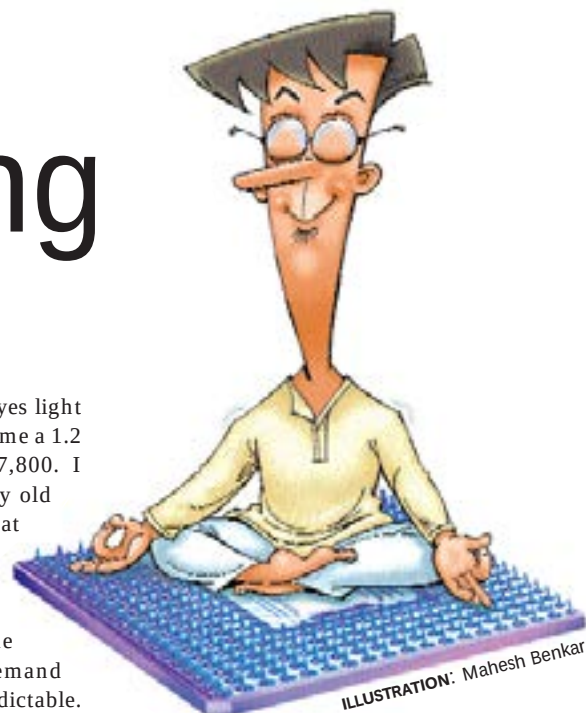
Tualatin processor. The dealer's eyes light up and he states that he can offer me a 1.2 GHz processor for around Rs 7,800. I inquire whether it can run on my old 815E motherboard, but am told that I would need to upgrade my board. Ironically these slower (and old generation) processors are more expensive than some of the P4 CPUs as there's little demand for them and availability is unpredictable. According to the dealer the P4 1.7 GHz (Socket 478), which costs Rs 7,350, is the most sensible choice, but if you're looking for something cheaper then a 1.1 GHz Celeron for Rs 5,800 is a good option.

The next shop is a genuine Intel dealer. He doesn't have the P4 1.6 GHz (Northwood) CPU in stock as it's not readily available—the processor can be bought for as little as Rs 8,500 depending on availability. The 1.9 GHz CPU is more easily available and at Rs 10,000 is the current hot seller. I ask for something below Rs 8,000 and he offers a Celeron 1 GHz (Tualatin) for just Rs 4,500. This processor works at a FSB of 100 MHz and requires an 815EP class motherboard to support it.

I move on to a new shop and ask for an older generation processor. The dealer has a couple of Duron 800s, which he's ready to sell for Rs 2,800. He even has an old P-III 1 GHz processor for Rs 8,500. It's way too expensive for an old generation P-III, but some people still prefer buying these CPUs for a dual processor workstation.

My final stop is Chunam Lane in Lamington Road, where I ask for a Duron processor. The dealer suggests that since I'm ready to spend Rs 8,000, I should opt for an Athlon XP instead. With the release of the Athlon XP 2100+, the prices of the lower-clocked CPUs have dropped substantially and an Athlon XP 1700+ is available for about Rs 7,500. I insist on a Duron, and he suggests the Duron 1.2 GHz processor, which retails at about Rs 6,000.

When upgrading from an Athlon Thunderbird core, ensure that your motherboard supports the new Palomino core—the same applies to the new Durons as



well. Also, AMD processors are now shipped in a retail box with an AMD recommended heatsink. Do not get conned into paying extra for the heatsink.

The Athlon XP hits the sweet spot, but you need DDR RAM to get maximum performance. But even a low-end CPU (like the 1600+) clubbed with SDRAM memory will pack a decent punch. The P4 1.6 GHz (Northwood) is another wonderful bargain, and when paired with DDR SDRAM, forms a potent combination.

Before you Upgrade...

- Ensure that your motherboard supports voltage and FSB requirements of the new CPU.
- If you're using a slot interface board then you might have trouble upgrading since socket adapters for recent processors such as the Athlon XP and the P-III (Tualatin) are not available.
- Older socket 423 motherboards can only support the slower P4s (1.4 to 1.8 GHz). You cannot run the latest P4s (2.0 GHz and above)
- If you have an old motherboard and are using a 200 MHz FSB-based Athlon CPU, then you will not be able to upgrade to the newer Athlon CPUs that use a 266 MHz FSB.
- If you are planning to upgrade to a P4 or an Athlon XP, then use RDRAM memory for the P4 or DDR SDRAM for the Athlon XP. Most new applications and games need a lot of RAM and 256 MB should be considered as the absolute minimum.



Shopping for Writers

With the release of 32x and 40x models, the prices of 24x CD-RW drives have crashed below the Rs 5,000 mark. You can get a Samsung 24x writer for approximately Rs 4,500, while the ASUS 24x unit costs Rs 5,000!